ML-Driven Traffic Flow Prediction using Fourier and Matrix-Based Features

group-9

CB.SC.U4AIE24119-Kiranmai
CB.SC.U4AIE24120-Sudeshna
CB.SC.U4AIE24143-P. Myagi Ranganadh
CB.SC.U4AIE24163-Devisri

INTRODUCTION

- Traffic congestion is a major issue in urban areas, leading to delays, fuel consumption, and environmental pollution. To address this problem, this project focuses on traffic flow analysis and prediction using machine learning techniques.
- Traditional traffic monitoring methods often struggle to effectively analyze large volumes of traffic data and identify hidden patterns.

OBJECTIVES

- To develop an ML-based traffic prediction system using historical traffic data.
- To model short-term traffic dependencies using Toeplitz-based time-series features.
- To analyze periodic and repetitive traffic patterns using FFT-based frequency features.
- To predict total traffic volume, individual vehicle counts, and traffic congestion levels.

METHODOLOGY

1. Data Collection

- Collect historical traffic data including:
  - Time, Day of the Week
  - Vehicle counts: CarCount, BikeCount, BusCount, TruckCount
  - Traffic Situation (Low, Medium, Heavy)

2. Feature Engineering

- Extract Hour from Time
- Encode Day numerically using Label Encoding:
  - Day_encoded=LabelEncoder(Day)
- Encode Travel Congestion numerically using Label Encoding:
  - Traffic_encoded=LabelEncoder(Traffic)

3. TOEPLITZ MATRIX

- Capture historical traffic patterns as features for prediction.
- Here, k=3 we are taking current and past two values.
- if k=1:
  - Each row only has the current timestep (t).
  - No history is captured .
  - Model sees only “now,” cannot learn trends from the past.
- if k =2:
  - Each row has current + previous timestep (t,t-1).
  - Captures very short-term trends.
- if k =10:
  - Each row contains current + last 9 timesteps(t,t-1,.....,t-9).
  - More features → larger feature matrix → slower training.
  - Risk of overfitting (model might memorize history).
- We can take k values as 3 or 4, but we took here as 3.

Example,

signal = [8,10,12,15,14,13]

- Each row consists of 3 values, as k =3.
- n = len(signal)=6
- First row starts at index k-1 = 2 (third element, 12) because we need 2 previous values. As first and second row won't have past values.

$$T[i,j] = \begin{cases} \text{signal}[i-j], & i-j \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

where, i = 2 to n

j = 0,1,2

Toeplitz matrix is

$$\begin{bmatrix} 12 & 10 & 8 \\ 15 & 12 & 10 \\ 14 & 15 & 12 \\ 13 & 14 & 15 \end{bmatrix}$$

- We are also calculating toeplitz matrix for toe_car, toe_bus, toe_bike ,toe_truck.
- We are combining all this matrices as toeplitz_features

3. FFT (Fast Fourier Transform)

- Fast Fourier Transform (FFT) is used to convert time-domain data into frequency-domain data to analyze repeating traffic patterns.
- The first few FFT components represent the lowest frequencies, which capture the main traffic trends and dominant cycles.
- Lower frequency components represent long-term trends in the data, such as gradual increases or decreases in traffic flow.
- These trends capture the overall behavior of the system, which is more stable and consistent over time.
- Machine learning models learn patterns more effectively when the input features contain stable and meaningful information rather than random fluctuations.
- Higher frequency components mainly represent rapid fluctuations or noise, which are less useful for prediction models.

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$$

Example , signal = [10,20,30,10]

- N = 4
- k = 0,1,2,3

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$$

- X[0]
  - $$X[0] = 10e^{-j2\pi(0)(0)/4} + 20e^{-j2\pi(0)(1)/4} + 30e^{-j2\pi(0)(2)/4} + 10e^{-j2\pi(0)(3)/4}$$

    X[0] = 10 + 20+ 30 +40 = 70
- X[1]
  - $$X[1] = 10e^{-j2\pi(1)(0)/4} + 20e^{-j2\pi(1)(1)/4} + 30e^{-j2\pi(1)(2)/4} + 10e^{-j2\pi(1)(3)/4}$$

    X[1] = 10 + (-20j) - 30 +10j = -10j - 20

    = abs(-10j -20) = 22.36
- X[2]
  - $$X[2] = 10e^{-j2\pi(2)(0)/4} + 20e^{-j2\pi(2)(1)/4} + 30e^{-j2\pi(2)(2)/4} + 10e^{-j2\pi(2)(3)/4}$$

    X[2] = 10 + (-20) + 30 -10

    = 10

- X[3]
  - $$X[3] = 10e^{-j2\pi(3)(0)/4} + 20e^{-j2\pi(3)(1)/4} + 30e^{-j2\pi(3)(2)/4} + 10e^{-j2\pi(3)(3)/4}$$

    X[3] = 10 + 20j -30 -10j = -20 +10j

    = abs(-20 +10j) = 22.36

These are the fft features = [70,22.36,10,22.36]

4. Feature Matrix & Targets

- Form Features (X):
  - Hour, Day_encoded, bike Count, car count, bus count, truck count,
  - toeplitz_ features, fft_all.
- Form Targets (y):
  - Regression: Total traffic, individual vehicle counts
  - Classification: Traffic situation
- Here, we are predicting total no of vehicles,bike count, car count, bus count, truck count separately.
  - For predicting bike count, we will exclude bike count column from feature matrix (X).
  - As they are ‘y’ which are target values.
  - If we include it in features, it will leads to data leakage.
- As same is done for other vehicles.
- Final size of Feature Matrix, n_samplesx30 for prediction of total no of vehicles.
- n_samplesx29 is for separate vehicles.

5. Train-Test Split

- Split data into training (80%) and testing (20%) sets

6. Decision Tree :

- Each internal node represents a feature to split on.
- Each branch represents a decision rule (threshold).
- Each leaf node represents the output value (prediction).
- We need the best_feature and best_threshold to know best_split.
- For each split, we use features (in random).
- To get best split, we need score that is impurity for regression.
- Task = regression ,

$$\text{Impurity} = n_{\text{left}} \cdot \text{Var}(y_{\text{left}}) + n_{\text{right}} \cdot \text{Var}(y_{\text{right}})$$

$$\text{Var}(y) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2$$

n left, n right = no of sample in left, no of sample in right

- To get best split for traffic congestion, we need score that is gini for classification.
- Task = classification ,

$$\text{Impurity} = n_{\text{left}} \cdot Gini(y_{\text{left}}) + n_{\text{right}} \cdot Gini(y_{\text{right}})$$

$$Gini = 1 - \sum_{k=1}^{C} p_k^2$$

k = number of classes

pk = how many times a class occured / total no of classes.

- By doing best split for each level , we built a Decision Tree.
- For each tree, we are doing bootstrap sampling(random sampling) where we can take each sample more than once.
- We are builting 15 decision trees to form a Random Forest.

- Now we have 15 trees.
- For regression,
  - we take mean for all the trees

$$\hat{y} = \frac{1}{N_{\text{trees}}} \sum_{i=1}^{N_{\text{trees}}} y_i$$

- For classification in traffic congestion,
  - we take mode method.

$$\hat{y} = \text{mode}\{y_1, y_2, \ldots, y_{N_{\text{trees}}}\}$$

7. Prediction

- Predict traffic values for test set:
  - Total traffic
  - Individual counts: Car, Bike, Bus, Truck
  - Traffic situation (Light / Medium / High/Heavy)

- As same as prediction of total no of vehicles, we do for separate vehicles as car_pred, bike_pred,bus_pred,truck_pred.
- And we are also predicting future 10 values for separate vehicles and also total vehicles.
- Same method is used for prediction of future values.

7. Evaluation

- Regression Metrics:
  - Mean Absolute Error (MAE):

$$MAE = \frac{1}{n}\sum_{i=1}^{n} | y_i - \hat{y}_i |$$

  - Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

- Classification Metrics:
  - Accuracy formula: how many predictions are completely correct out of all predictions.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

  - Precision formula: Of all predictions for a class, how many were correct

$$\text{Precision} = \frac{TP}{TP + FP}$$

  - Recall formula: Of all actual samples for a class, how many did we correctly predict

$$\text{Recall} = \frac{TP}{TP + FN}$$

- F1-score Formula:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

8. Visualization

- Compare Actual vs Predicted for the last 10 samples:
  - Total traffic
  - Individual vehicle counts
  - Traffic situation

RESULTS

```
--- Regression (Total Vehicles) ---
MAE : 5.307278293599404
RMSE: 6.957378190898284
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--- Classification (Traffic) ---
Accuracy : 0.957983193277311
Precision: 0.9576001974107966
Recall   : 0.957983193277311
F1-score : 0.9576696667082993
```

--- Last 10 Predictions ---

	Actual_Bike	Pred_Bike	Actual_Car	Pred_Car	Actual_Bus	Pred_Bus	Actual_Truck	Pred_Truck	Actual_Total	Pred_Total	Actual_Traffic	Pred_Traffic
0	25	23	97	92	19	17	24	21	165	154	high	high
1	18	19	101	99	3	7	17	18	139	141	normal	normal
2	22	21	129	115	42	25	1	1	194	178	heavy	heavy
3	22	20	42	48	5	12	9	10	78	78	low	low
4	24	18	18	41	21	23	22	21	85	90	normal	normal
5	0	1	12	12	1	1	34	30	47	46	normal	normal
6	17	18	123	122	50	36	1	1	191	191	heavy	heavy
7	27	27	123	124	22	26	4	2	176	182	heavy	heavy
8	5	3	13	14	0	1	17	18	35	36	normal	normal
9	12	16	102	115	24	28	3	2	141	160	normal	normal

THANK YOU